Think — nets in context

Designing a real box: the same six faces, laid out two ways, and two very different bills for card.

LESSON 190 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 8.4 IN CONTEXT

LESSON CLOCK

10' | 12' | 12' | 6'

The task	10 min
Two layouts for the same box	12 min
Waste, tabs and the sheet	12 min
Back to the shape of the box	6 min

LEARNING OBJECTIVES

- Design a net for a box of given dimensions, with tabs.
- Find the smallest rectangle of card the net will fit into.
- Compare two layouts by the card each wastes.
- Relate the result back to the shape of the box itself.

TEXTBOOK

National	pp. 542–544
Cambridge	Stage 7, pp. 94–96
Workbook	Investigation 8

01 Explanation

The task

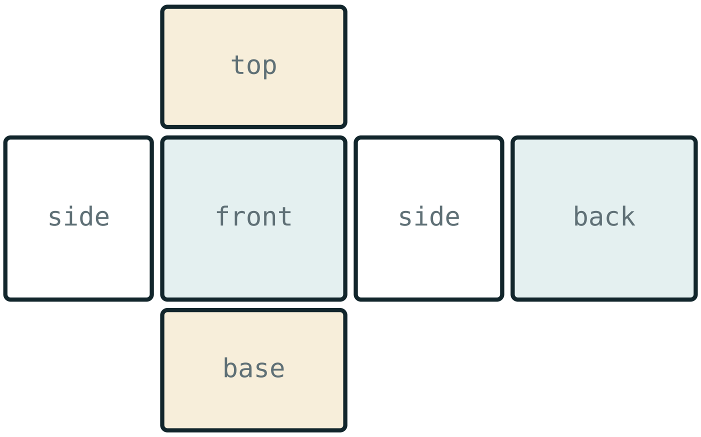
TODAY’S TASK

A box must hold *10* cm by *8* cm by *5* cm of tea. Design its net, decide how it should be laid out on a sheet of card, and say how much card is actually bought for each box.

QUANTITY	WORKING	VALUE
volume	$10 \cdot 8 \cdot 5$	400 cm³
the three faces	$80, 40, 50$	cm²
surface area	$2(80 + 40 + 50)$	340 cm²

The *340* cm² is the card that ends up as box. It is not the card that is bought.

Two layouts for the same box



A net laid out flat — the pieces are fixed, their arrangement is not

LAYOUT	ARRANGEMENT	SMALLEST RECTANGLE	ITS AREA
the cross	a column of four faces with the two sides beside the front	*26 × 26 cm*	*676 cm²*
the row	four faces in a row, top above and bottom below	*36 × 21 cm*	*756 cm²*

WHERE THE *26*S COME FROM

Down the column: $8 + 5 + 8 + 5 = 26$ cm. Across: $8 + 10 + 8 = 26$ cm. The row layout is $10 + 8 + 10 + 8 = 36$ cm across and $8 + 5 + 8 = 21$ cm down.

Waste, tabs and the sheet

LAYOUT	CARD BOUGHT	CARD USED	WASTE	WASTE AS A SHARE
the cross	*676 cm²*	*340 cm²*	*336 cm²*	about *50*%
the row	*756 cm²*	*340 cm²*	*416 cm²*	about *55*%

Tabs are extra. Seven glued edges of total length *46* cm, each given a *1* cm tab, add *46* cm² more card — and they must fit inside the same rectangle.

THE SURFACE AREA IS THE SAME IN BOTH LAYOUTS

Both boxes use *340* cm² of card as box. What differs is the rectangle it has to be cut from, and the offcuts are paid for. On a run of *10 000* boxes the cross layout saves *80* cm² each — *80* m² of card.

Back to the shape of the box

Project 3 of Quarter III found that, for a fixed volume, the cube uses least card. This box holds *400* cm³, so the cube would have edge *7.4* cm.

BOX	VOLUME	SURFACE AREA
$10 \times 8 \times 5$	*400 cm³*	*340 cm²*
cube of edge *7.4*	*about 400 cm³*	about *326* cm²

TWO SAVINGS, OF VERY DIFFERENT SIZES

Reshaping the box towards a cube saves about *14* cm²; laying the net out well saves *80* cm². For this box the layout matters more than the shape — which is not what most people guess, and is exactly the kind of thing a report should say.

02 Worked examples

EXAMPLE 1 **Find the card used, and the card bought, for the cross layout.**

1. Used: $2(80 + 40 + 50) = 340$ cm².
2. Bought: the rectangle *26* by *26* cm.
3. *676* cm², so *336* cm² is waste.
   About half of what is bought ✓

ANSWER

340 used, *676* bought

EXAMPLE 2 **How much card does the cross layout save over the row layout on *10 000* boxes?**

1. $756 - 676 = 80$ cm² a box.

2. $80 \cdot 10\,000 = 800\,000$ cm².

3. $800\,000 \div 10\,000 = 80$ m².

   1 m² is $10\,000$ cm².

ANSWER

80 m²

EXAMPLE 3 **A cube holding the same *400* cm³ has edge about *7.4* cm. Find its surface area and compare.**

1. $7.4^2 = 54.76$ cm² a face.

2. $6 \cdot 54.76 \approx 326$ cm².

3. $340 - 326 = 14$ cm² saved.

   Less than the 80 cm² the layout saves.

ANSWER

About 326 cm² — a saving of 14 cm²

03 Interactive model

Let pairs cut both layouts from squared paper and measure the offcuts; the “obvious” row layout losing to the cross is the moment worth waiting for.

Card used and card bought

A $10 \times 8 \times 5$ cm box has surface area:

- 170
- 340
- 400
- 676

Its volume is:

- 170
- 340
- 400
- 676

The cross layout fits into a rectangle:

- 26×26
- 36×21
- 34×20
- 26×36

The row layout fits into:

- 26×26
- 36×21
- 21×21
- 36×26

The cross layout wastes:

- 80
- 336
- 416
- 676

The cross saves, per box:

- 14 cm²
- 80 cm²
- 336 cm²
- nothing

A cube of the same volume saves about:

- 14 cm²
- 80 cm²
- 340 cm²
- nothing

For this box the bigger saving comes from:

- the shape
- the layout
- the tabs
- the colour

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Layout	Joylashtirish	Раскладка
Tab (flap)	Yopishtirish qismi	Клапан
Waste	Chiqindi	Отходы
Offcut	Qirqindi	Обрезок
Bounding rectangle	O‘rovchi to‘rtburchak	Охватывающий прямоугольник
To nest	Zich joylashtirmoq	Компоновать
Efficiency	Samaradorlik	Эффективность
Production run	Ishlab chiqarish partiyasi	Партия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The card used as box is:

- the bounding rectangle
- the surface area
- the volume
- the waste

The card bought is:

- the surface area
- the smallest rectangle the net fits into
- the volume
- the tabs

Changing the layout changes:

- the surface area
- the waste
- the volume
- the number of faces

Tabs are needed so that:

- the box looks better
- the faces can be glued
- the volume grows
- the net is square

Nesting the nets of many boxes:

- wastes more
- shares the offcuts
- changes the volume
- needs no tabs

A good report gives:

- only the best answer
- the comparison and its limits
- the volume alone
- a drawing only

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The surface area of a $10 \times 8 \times 5$ cm box

   ANSWER 340 cm²

2. Its volume

   ANSWER 400 cm³

3. The faces in its net

   ANSWER 6

4. Tabs are needed so that

   ANSWER The faces can be glued together

5. The bounding rectangle of the cross layout

   ANSWER 26 by 26 cm

6. Its area

   ANSWER 676 cm²

7. The card that ends up as box

   ANSWER 340 cm²

Medium · the standard question

7 PROBLEMS

1. The waste in the cross layout

   ANSWER 336 cm²

2. The bounding rectangle of the row layout

   ANSWER 36 by 21 cm

3. Its area

   ANSWER 756 cm²

4. The waste in the row layout

   ANSWER 416 cm²

5. Which layout wastes less

   ANSWER The cross

6. The waste of the cross layout as a share

   ANSWER About 50%

7. The waste of the row layout as a share

   ANSWER About 55%

Hard · stretch and reason

7 PROBLEMS

1. A cube holding *400* cm³ has edge, to 1 d.p.

   ANSWER *7.4* cm

2. Its surface area

   ANSWER About *326* cm²

3. The card a cube would save

   ANSWER About *14* cm² a box

4. The card the better layout saves

   ANSWER *80* cm² a box

5. On *10 000* boxes, that saving

   ANSWER *80* m²

6. *1* cm tabs on seven edges totalling *46* cm add

   ANSWER *46* cm²

7. Why does the layout matter if the surface area is fixed?

   ANSWER The offcuts of the rectangle are paid for too

07 Homework

Homework — the task

One page: the net with its measurements, the two layouts, the waste of each, and one conclusion.

1. Draw a net for a *12* cm by *6* cm by *4* cm box and find its surface area.
2. Lay that net out two different ways and give the smallest rectangle each fits into.
3. Work out the waste of each layout, in cm² and as a percentage.
4. Find the edge and the surface area of a cube holding the same volume.
5. Write one sentence saying whether the shape or the layout saves more card for your box.